

Engineering——计数器级联

Cycle

1.design

在 3-Bit 基因逻辑计数器的设计初期，系统在静态等待阶段极易因上游基底表达积累而出现高位bit2提前翻转的“早期泄露”问题。为此，我们打破单一前馈环的拓扑限制，提出了重构级联逻辑门与优化门响应特性的双重策略：一是引入结合外源时钟信号 Int_0 与上游进位信号的双因子杂合 AND 门，强制要求双信号同时存在；二是意识到单纯调节 mRNA 或蛋白成熟等表达时间无法解决泄露，必须优先提高门的状态分辨能力，即响应陡峭度。

2.build

我们将 Pokhilko 重组动力学与杂合逻辑门网络相结合，建立了描述 DNA 拓扑状态与转录调控动态的非线性 ODE 模型。我们在模型中引入了独立的门响应指数 $n_{A1,gate}$ 并结合时钟 AND 门控。相比于无法明显抑泄露的 A1单因子负自馈，杂合逻辑 AND 门成功阻断了静态等待期间因单一上游积累导致的非预期表达；同时，扫描结果证实将门指数提升至 $n = 6$ 能大幅降低泄露指标——较 $n = 4$ 降低约 88%，效果远优于单纯调整表达时间。

3.test

数值仿真结果显示，在引入杂合逻辑门并优化门响应特性后，系统成功实现了从 P1 (0, 0, 1) 到 P8 (0, 0, 0) 的 8 阶段完整二进制递增计数与循环复位，波形陡峭且具备明确的时序裕量。然而，在参数波动和敏感性检验下，部分 Bit 的提前翻转仍难以完全避免，表明单一门的陡峭响应仍难以完全抵御全网参数扰动，对参数窗口存在一定依赖。

4.learn

从本次迭代中我们认识到，解决复杂的动态泄露不能拘泥于初始的简单前馈环拓扑，也不能依赖在响应分离度不足的元件上寻找狭窄的时间窗口。工程设计应当确立明确的优先级策略：先通过拓扑重构与元件筛选获得足够的开关分离度，再用表达动力学完成时序匹配。这一认识为后续从模块接口要求出发筛选高陡峭度元件、构建更加稳健的基因逻辑计数器奠定了坚实的基础。

CYCLE 1

1.design

In the early stages of designing the 3-bit genetic counter, the system suffered from "early leakage," where higher-order bits prematurely flipped during static waiting phases due to upstream basal expression accumulation. To overcome the topological constraints of simple feed-forward loops, we proposed a dual strategy combining logic gate reconstruction and gate response optimization: (1) implementing a dual-input hybrid AND gate requiring both the upstream carry signal and the external clock signal Int_0 , and (2) recognizing that tuning expression kinetics, like mRNA and protein maturity, alone cannot resolve leakage, prioritizing higher state-resolving power, namely gate response steepness.

2.build

We integrated Pokhilko recombination dynamics with the hybrid logic gate network to construct a non-linear ODE model. We introduced an independent gate response coefficient $n_{A1,gate}$ alongside clock AND gating. Unlike single-factor A_1 negative autoregulation—which failed to yield significant leakage suppression—the hybrid AND gate successfully blocked unintended expression during waiting periods. Furthermore, parameter scanning confirmed that increasing $n_{A1,gate}$ to 6 reduced leakage by 88% compared to $n = 4$, far outperforming expression delay adjustments alone.

3.test

Numerical simulations showed that after integrating hybrid logic gates and optimizing response characteristics, the system successfully accomplished an 8-stage binary increment from P1 (0, 0, 1) to P8 (0, 0, 0) with cyclic resetting, sharp transitions, and clear timing margins. However, under parameter fluctuations and sensitivity testing, premature bit flipping could not be entirely prevented, indicating that single-gate steepness alone is insufficient to fully buffer against network-wide perturbations, leaving a dependency on the parameter window.

4.learn

From this iteration, we learned that resolving complex dynamic leakage cannot rely on initial simple feed-forward topologies or searching for narrow time windows among components with poor state separation. Engineering design must establish a clear priority rule: First achieve sufficient ON/OFF separation via topological reconstruction and component screening, then align temporal kinetics through expression dynamics. This insight provides a solid foundation for screening high-steepness components based on module interface requirements to construct a more robust genetic logic counter.